Cover Letter

The Editor
Nature Physics
Springer Nature
Date: May 15, 2025

Re: Submission of original research article - "Collapse-Originated Quantum Gravity: A Prime-Resonant Unification of Time, Space, and Information"

Dear Editor,

I am pleased to submit our manuscript titled "Collapse-Originated Quantum Gravity: A Prime-Resonant Unification of Time, Space, and Information" for consideration as an Article in Nature Physics.

This work presents a novel quantum gravity framework that fundamentally reframes our understanding of gravitational phenomena. Rather than treating gravity as a geometric force or a quantizable field, we derive it from structural recursion embedded in collapse processes occurring at prime-indexed attractor points. Our approach offers a mathematical foundation that naturally addresses long-standing tensions between quantum mechanics and general relativity.

The key innovations of our work include:

1. A unified evolution equation that generates gravitational dynamics through self-referential collapse and golden-ratio modulated structures
2. A mathematically rigorous coupling between information density and spacetime curvature
3. Testable predictions for black hole information interference and gravitational wave subharmonic structures
4. A first-principles derivation of gravity as an emergent feature of collapse dynamics, without introducing additional dimensions or entities

Our findings align with Nature Physics' focus on groundbreaking research that transforms our understanding of fundamental physical principles. The mathematical framework we present has implications across multiple domains, including quantum foundations, gravitational theory, cosmology, and information physics.

We believe this work is particularly suitable for Nature Physics for several reasons:

1. **Interdisciplinary significance**: Our framework bridges quantum information theory, gravitational physics, and number theory in a novel way that will interest your diverse readership spanning multiple physics specialties.

2. **Experimental testability**: Unlike many quantum gravity proposals, our model makes specific predictions about gravitational wave subhar-

monics and black hole information patterns that can be tested with next-generation LIGO/Virgo observations and Event Horizon Telescope data, aligning with your journal's emphasis on connecting theory with experimental validation.

3. **Conceptual breakthrough**: Our prime-resonant approach fundamentally reframes the nature of spacetime as emergent from information-theoretic collapse, offering a genuinely new perspective on one of physics' most persistent challenges.

4. **Mathematical elegance**: The golden-ratio modulated structures and prime-indexed attractors provide a mathematically cohesive foundation that unifies previously disparate aspects of quantum theory and gravity.

The potential impact of this work extends beyond the theoretical physics community. The information-theoretic foundation may provide new computational approaches to gravitational simulation, while the prime-indexed attractor concept offers novel mathematical tools for analyzing complex systems across disciplines. Our collapse-originated model also suggests new interpretations of cosmological observations, potentially reshaping our understanding of dark energy and the early universe.

The manuscript has not been previously published, nor is it under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission to Nature Physics. We have no conflicts of interest to disclose.

We suggest the following scientists as potential reviewers:

1. Prof. Carlo Rovelli, Aix-Marseille University - Pioneer in loop quantum gravity and relational quantum mechanics
2. Prof. Abhay Ashtekar, Pennsylvania State University - Founder of loop quantum gravity and expert in quantum geometry
3. Prof. Juan Maldacena, Institute for Advanced Study - Authority on AdS/CFT correspondence and holographic principles
4. Prof. Francesca Vidotto, Western University - Specialist in quantum cosmology and Planck-scale physics
5. Prof. Gerard 't Hooft, Utrecht University - Nobel laureate with expertise in black hole information paradox

We would also like to note that Prof. Leonard Susskind (Stanford University) and Prof. Robbert Dijkgraaf (Institute for Advanced Study) may have potential conflicts of interest due to their closely related theoretical work, though we welcome their expert assessment if you deem appropriate.

We sincerely appreciate your consideration of our manuscript and look forward to your response.

Respectfully,

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